

Azure Data Factory Data Transformation Using Filters & Joins

SQL Database → Mapping Data Flow → Azure Blob Storage

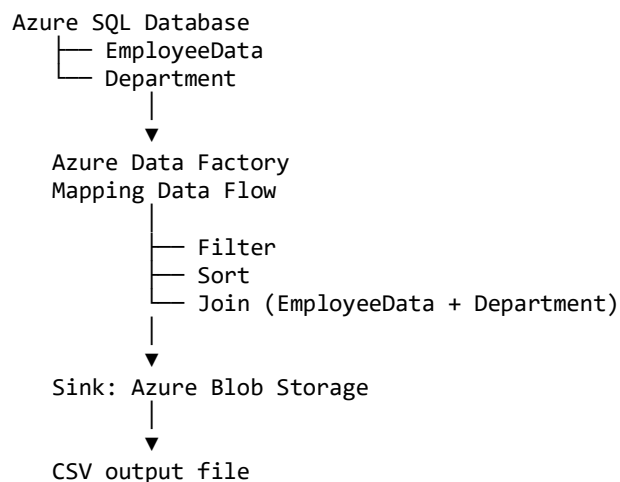
Practical step-by-step guide using EmployeeData and Department tables

1. What this guide covers

This guide demonstrates how to take the EmployeeData table from Azure SQL Database into Azure Data Factory, transform it using Mapping Data Flow, filter and sort the rows, and write the result to Azure Blob Storage. A second data flow shows how to bring EmployeeData and Department into ADF and join them using DepartmentID.

- Create/verify the Azure SQL source and Blob Storage destination.
- Create linked services and datasets in Azure Data Factory.
- Create a Mapping Data Flow with EmployeeData as the source.
- Apply a Filter transformation.
- Apply a Sort transformation.
- Write the transformed output to Blob Storage using a Sink.
- Create a second source for Department and combine the two streams using Join.
- Understand Inner, Left Outer, Right Outer and Full Outer joins.
- Debug, preview, validate and monitor the data flow.

2. Overall architecture



Mapping Data Flow provides a visual transformation experience. Data flows can contain one or more sources, transformation steps and a sink. Microsoft documents Filter as a row-level transformation and Join as a transformation that combines two streams.

3. Screenshots from the uploaded recordings

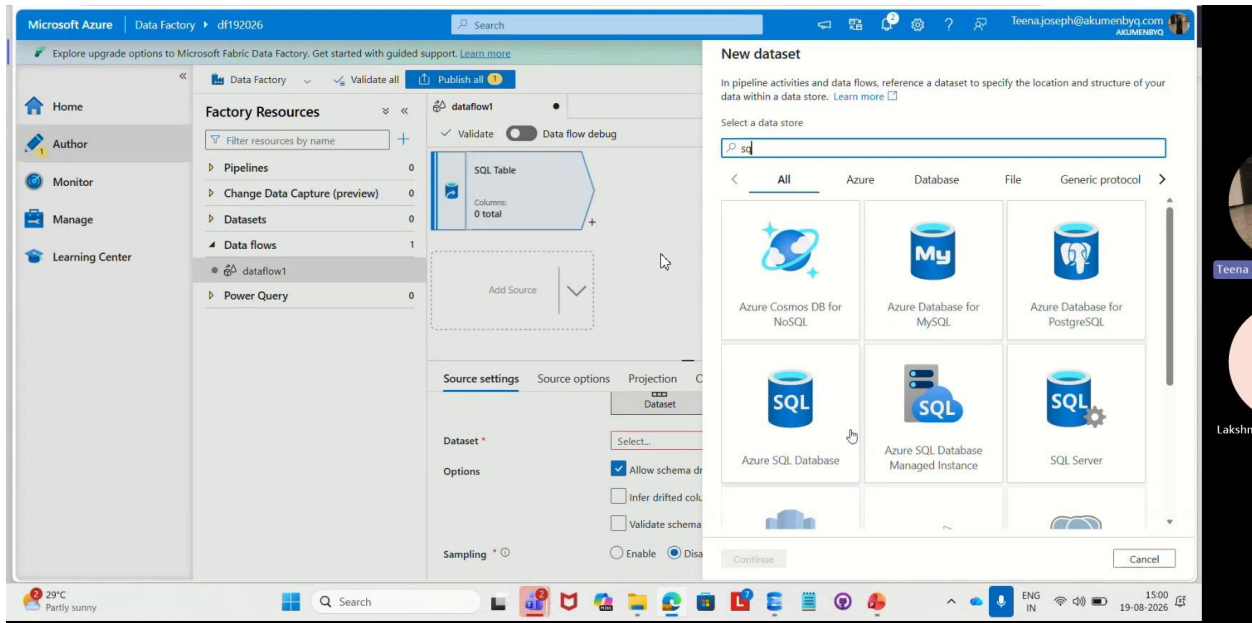


Figure 1 — Uploaded recording: ADF Data Flow canvas and source setup.

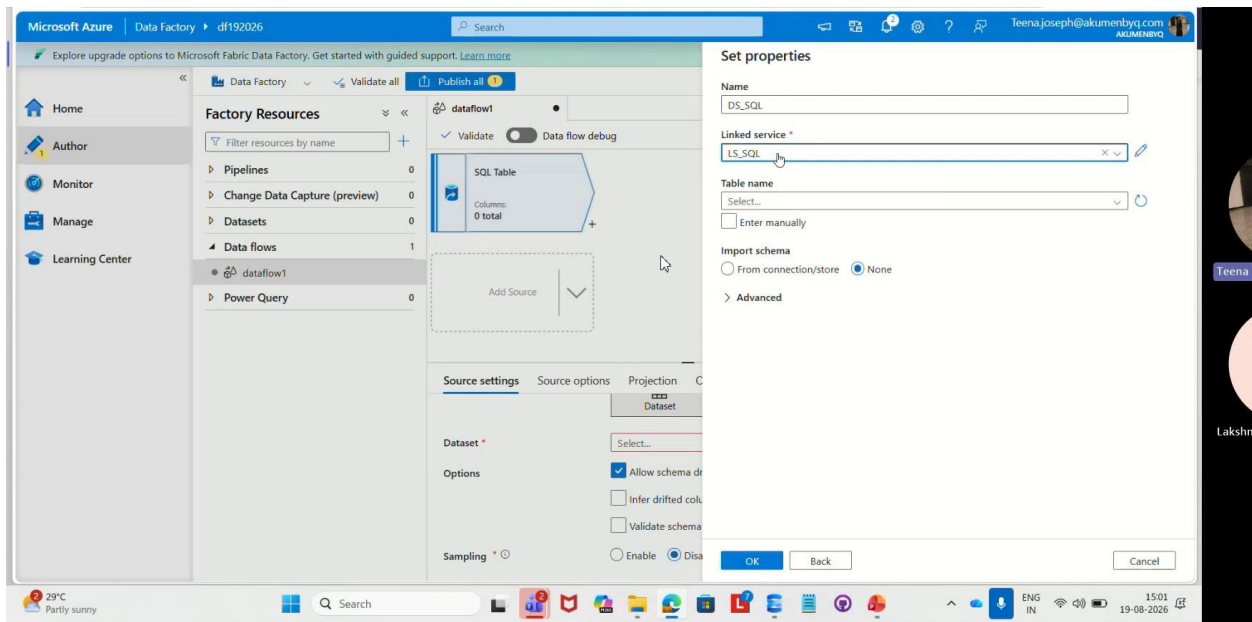


Figure 2 — Uploaded recording: configuring the SQL source in the Mapping Data Flow.

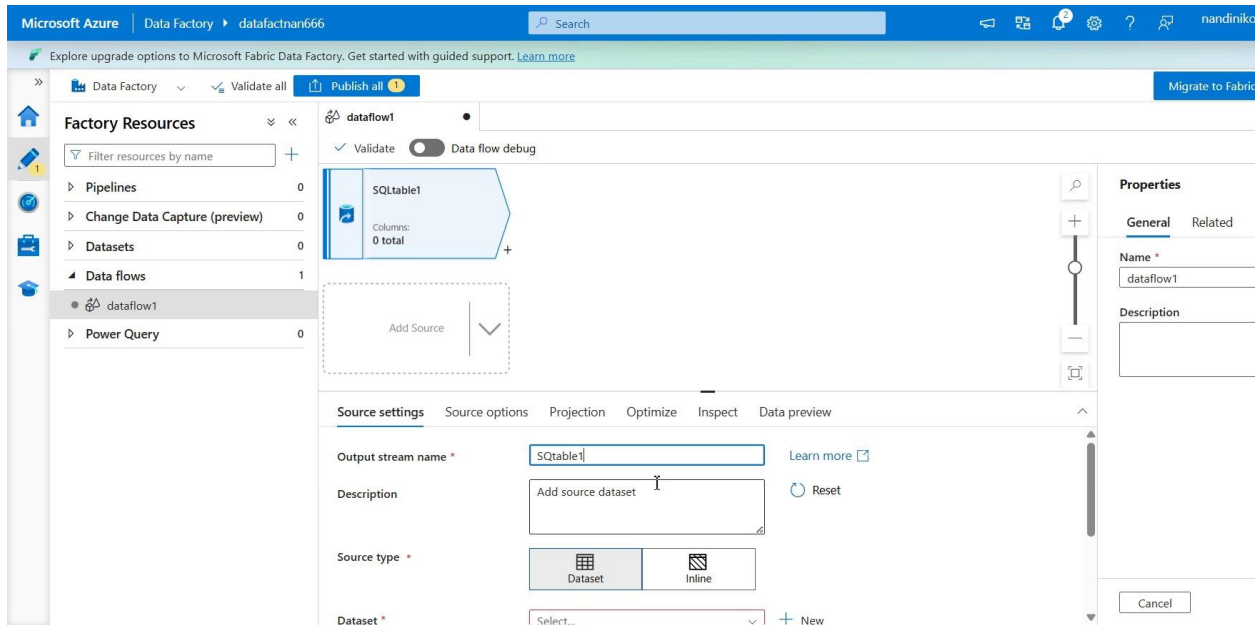


Figure 3 — Uploaded recording: source options/schema configuration area.

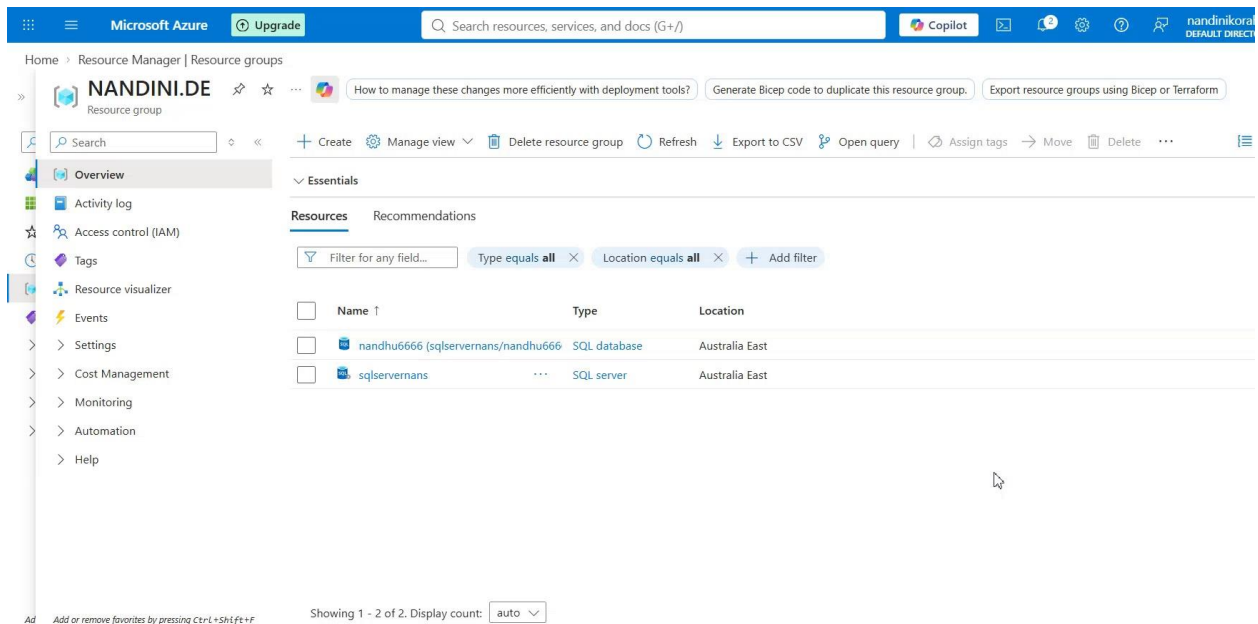


Figure 4 — Uploaded recording: Azure Storage account area.

4. Prerequisites

- An Azure subscription.
- An Azure SQL Database containing EmployeeData and Department.
- An Azure Storage Account with a Blob container, for example: adfoutput.
- An Azure Data Factory resource.
- SQL access credentials or Microsoft Entra authentication with permission to read the source tables.
- A Data Factory Integration Runtime. For Azure SQL Database and Azure Blob Storage, the Azure Integration Runtime is commonly used.

The screenshot shows the 'Create a storage account' page in the Azure portal. The breadcrumb trail is: Home > Resource Manager | Resource groups > NANDINI.DE > Marketplace > Storage account. The page title is 'Create a storage account'. The form includes the following fields and options:

- Resource group ***: NANDINI.DE (with a 'Create new' link).
- Storage account name ***: storeaccountnan666.
- Region ***: (US) East US (with a 'Deploy to an Azure Extended Zone' link).
- Primary service ***: Choose preferred storage type (with a dropdown arrow).
- Performance ***: ☒ Standard: Recommended for most scenarios (general-purpose v2 account). ☐ Premium: Recommended for scenarios that require low latency.
- Redundancy ***: Geo-redundant storage (GRS) (with a dropdown arrow). Below this, there are two checkboxes: ☒ Make read access to data available in the event of regional unavailability, and ☐ Geo priority replication guarantees Blob storage data is geo-replicated within 15 minutes.

At the bottom, there are three buttons: 'Previous', 'Next', and 'Review + create'.

Figure 5 — Uploaded recording: Create storage account screen.

The screenshot shows the 'Overview' page for the storage account 'storeaccountnan666_1787647671494'. The breadcrumb trail is: Home > storeaccountnan666_1787647671494 | Overview. The page title is 'Overview'. The page shows the following information:

- Deployment details**: Deployment name: storeaccountnan666_1787647671494, Subscription: Azure subscription 1, Resource group: NANDINI.DE, Start time: 8/25/2026, 2:17:56 PM, Correlation ID: f5b4368d-5d68-4edb-823c-2882c1dc2b15.
- Deployment details table**:

Resource	Type	Status	Operation details
storeaccountnan666	Microsoft.Storage/storageAccounts	Accepted	Operation details

At the bottom, there is a 'Give feedback' section with a link to 'Tell us about your experience with deployment'.

Figure 6 — Uploaded recording: storage account overview after creation.



SQL Database [Add to Favorites](#)

Microsoft | Azure Service

★ 4.4 (2537 ratings)

[Azure benefit eligible](#)

Subscription

Azure subscription 1

Plan

SQL Database

Create

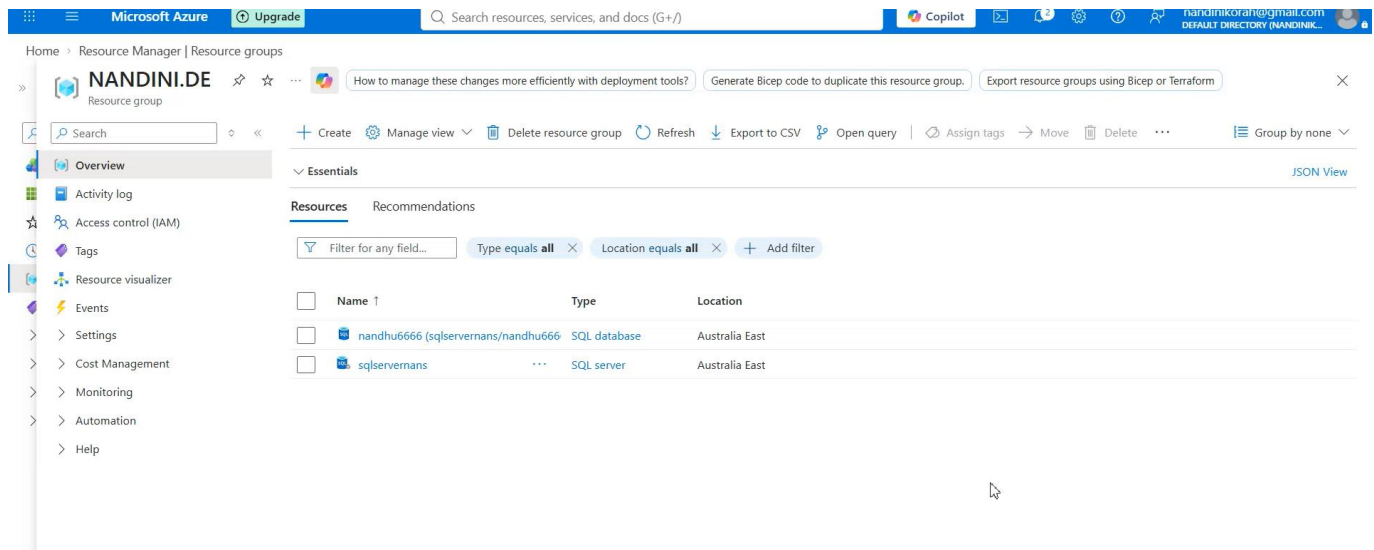
[Overview](#) [Plans](#) [Usage Information + Support](#) [Ratings + Reviews](#)

SQL Database is a cloud database service built for application developers that lets you scale on-the-fly without downtime and efficiently deliver your applications. Built-in advisors quickly learn your application's unique characteristics and dynamically adapt to maximize performance, reliability, and data protection.

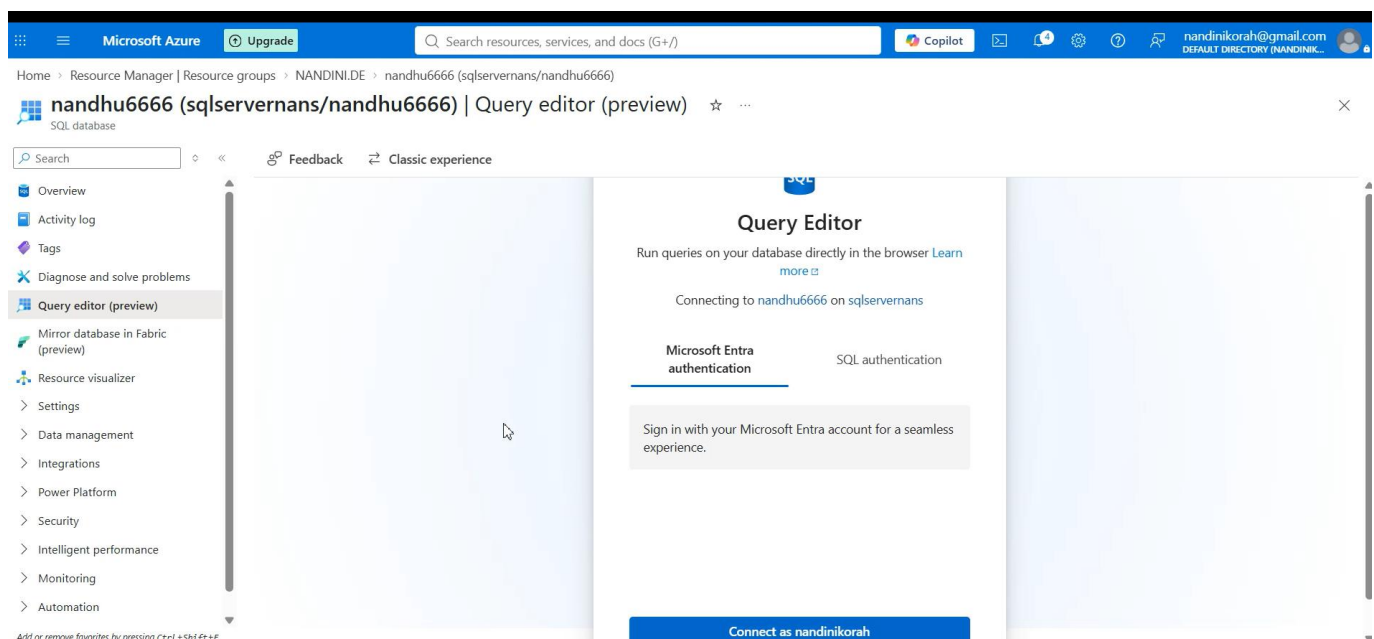
Use this template to create a new database in the SQL Database service. You can create the database on a new logical server or on a logical server that already exists in your subscription.

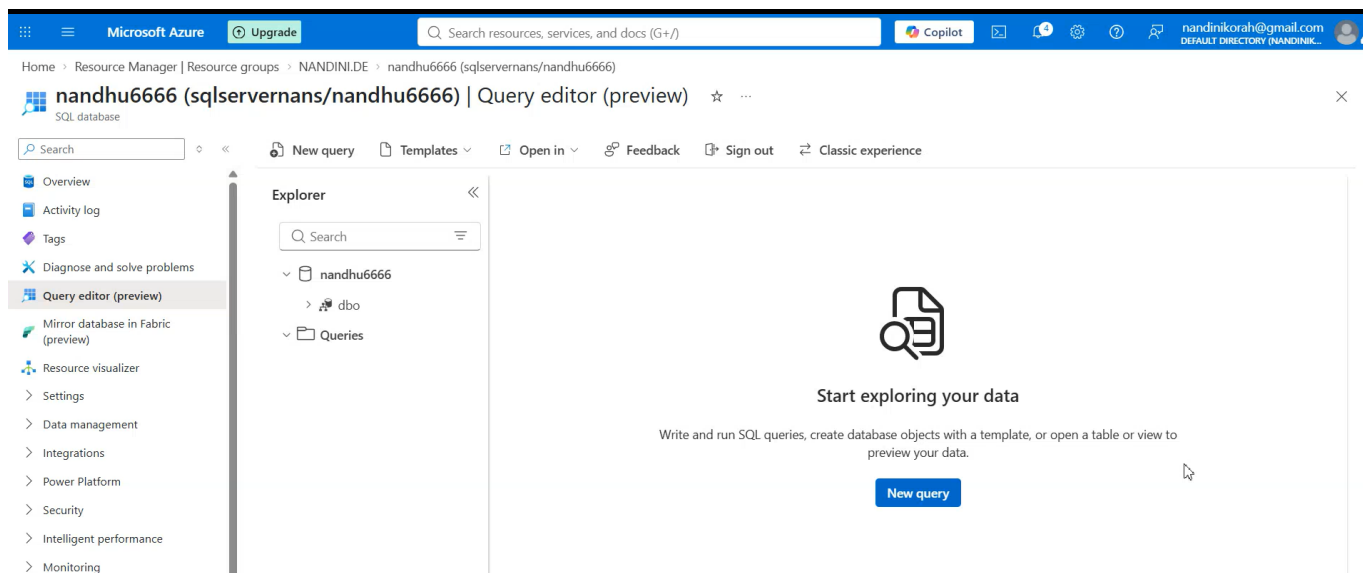
An Azure Data Factory resource.

The screenshot shows the Azure Data Factory console interface. At the top, the breadcrumb navigation reads 'Microsoft Azure | Data Factory | datafactnans666'. Below this is a search bar and a notification banner about upgrading to Microsoft Fabric Data Factory. A secondary banner provides information about configuring a Git repository. The main content area is titled 'Data factory datafactnans666' and includes a 'New' button. Four primary actions are highlighted in cards: 'Ingest' (Copy data at scale), 'Orchestrate' (Code-free data pipelines), 'Transform data' (Transform data using data flows), and 'Configure SSIS' (Manage & run your SSIS packages). At the bottom, a 'Recent resources' section is currently empty, displaying 'No items to show'.



SQL Database Creation





Suggested example table structures:

EmployeeData

EmployeeID	INT
EmployeeName	VARCHAR(100)
DepartmentID	INT
Salary	DECIMAL(10,2)
City	VARCHAR(50)

Department

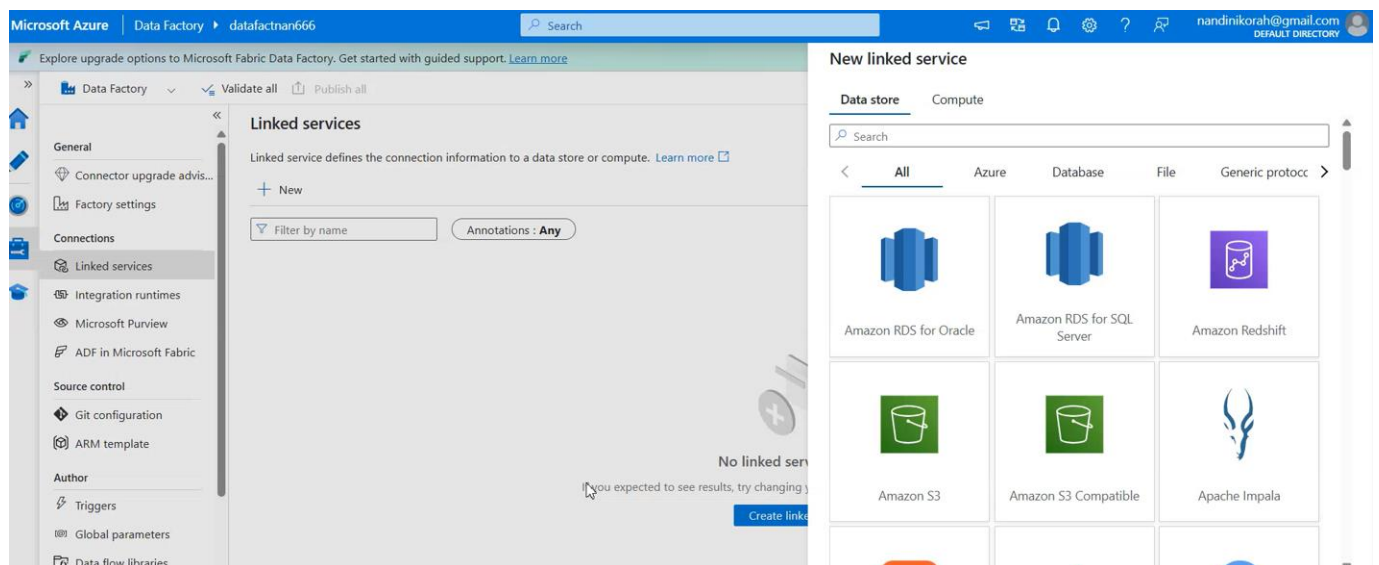
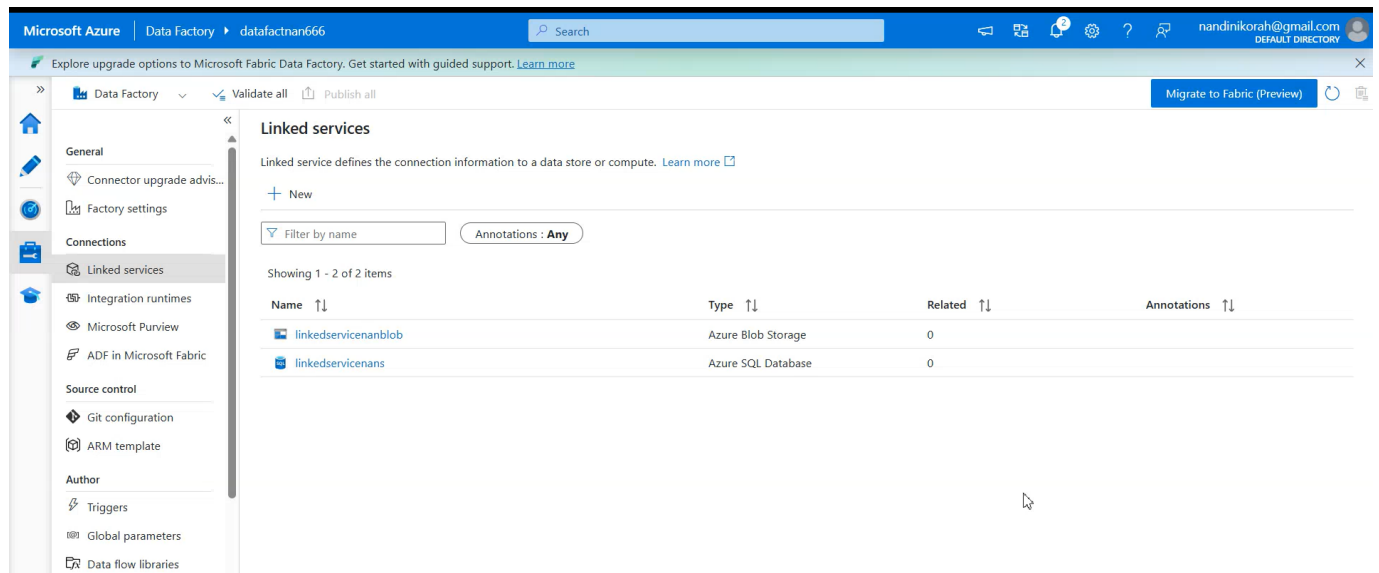
DepartmentID	INT
DepartmentName	VARCHAR(100)

The exact column names can be changed to match your database. In the examples below, DepartmentID is the join key.

5. Create the Linked Services and Datasets

5.1 Azure SQL Database linked service

- Open Azure Data Factory Studio.
- Select Manage → Linked services → + New.
- Choose Azure SQL Database.
- Give the linked service a name such as AzureSql_EmployeeDB.
- Select the Azure subscription, server and database.
- Choose the authentication method you use.
- Test the connection and create the linked service.



5.2 EmployeeData dataset

- Go to Author → Datasets → + New dataset.
- Choose Azure SQL Database.
- Select the Azure SQL linked service.
- Choose EmployeeData as the table.
- Name the dataset, for example DS_EmployeeData.
- Import/validate the schema.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Resource Manager | Resource groups > NANDINI.DE > nandhu6666 (sqlservnans/nandhu6666)

nandhu6666 (sqlservnans/nandhu6666) | Query editor (preview)

SQL database

Search

New query | Templates | Open in | Feedback | Sign out | Classic experience

Overview | Activity log | Tags | Diagnose and solve problems | **Query editor (preview)** | Mirror database in Fabric (preview) | Resource visualizer | Settings | Data management | Integrations | Power Platform | Security | Intelligent performance | Monitoring

Explorer

Search

- nandhu6666
 - dbo
 - Queries

Start exploring your data

Write and run SQL queries, create database objects with a template, or open a table or view to preview your data.

New query

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Resource Manager | Resource groups > NANDINI.DE > nandhu6666 (sqlservnans/nandhu6666)

nandhu6666 (sqlservnans/nandhu6666) | Query editor (preview)

SQL database

Search

New query | Templates | Open in | Feedback | Sign out | Classic experience

Overview | Activity log | Tags | Diagnose and solve problems | **Query editor (preview)** | Mirror database in Fabric (preview) | Resource visualizer | Settings | Data management | Integrations | Power Platform | Security | Intelligent performance | Monitoring

Explorer

SQL query 1

Run | Save as view

Your queries will be lost when your session ends. Use **Save as view** to keep results as a database view, or copy your query text to save it externally.

```

29 (23, 'Saijaya', 'Puri', 'Male', 43, 'Sales', 'Regional Sales Head', 75000, 10, 'Delhi', 103, '2000-06-06'),
30 (26, 'Julie', 'Dcruz', 'Female', 26, 'Marketing', 'Digital Marketer', 39000, 2, 'Mumbai', 104, '2021-01-25'),
31 (27, 'Rohan', 'Kulkarni', 'Male', 31, 'IT', 'DevOps Engineer', 61000, 6, 'Bangalore', 101, '2017-05-29'),
32 (28, 'Tanya', 'Shetty', 'Female', 34, 'Finance', 'Risk Analyst', 64000, 9, 'Pune', 102, '2015-10-10'),
33 (29, 'Akash', 'Mali', 'Male', 23, 'HR', 'HR Intern', 22000, 0, 'Mumbai', 105, '2024-01-10'),
34 (30, 'Reema', 'Jain', 'Female', 29, 'IT', 'QA Engineer', 50000, 5, 'Chennai', 101, '2019-12-19');
35 select * from EmployeeData;
  
```

Messages | Results | | | |

	EmpID	ABC FirstName	ABC LastName	ABC Gender	123 Age	ABC Departm...	ABC JobTitle	123 Salary	123 Experience	ABC City	123 ManagerID
1	1	Rahul	Sharma	Male	28	IT	Software Engi...	55000	3	Mumbai	101
2	2	Priya	Mehta	Female	32	HR	HR Manager	65000	7	Delhi	NULL
3	3	Amit	Verma	Male	45	Finance	Accountant	48000	15	Pune	102
4	4	Sneha	Patil	Female	29	IT	Data Analyst	52000	4	Bangalore	101
5	5	Karan	Singh	Male	35	Sales	Sales Executive	40000	8	Chennai	103

5.3 Department dataset

- Repeat the same process for the Department table.
- Name it DS_Department.
- Use the same Azure SQL linked service.

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > databasen66 (databaservernan66/databasen66)

databasen66 (databaservernan66/databasen66) | Query editor (preview) ☆

SQL database

Search

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance
Monitoring
Automation

Add or remove favorites by pressing Ctrl+Shift+F

New query Templates Open in Feedback Sign out Classic experience

SQL query 1

Run Save as view

Your queries will be lost when your session ends. Use **Save as view** to keep results as a database view, or copy your query text to save it externally.

```

38 );
39 INSERT INTO Department (DepartmentID, DepartmentName, Location)
40 VALUES
41 (1, 'HR', 'Kochi'),
42 (2, 'IT', 'Bangalore'),
43 (3, 'Finance', 'Chennai');

```

Messages Results

DeptID	DeptName	Location
1	HR	Kochi
2	IT	Bangalore
3	Finance	Chennai
4	Marketing	Mumbai
5	Sales	Delhi

Succeeded (0 sec 246 ms) Columns: 3 Rows: 5

nandini-sql-server EmployeeDB

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

Query 3

Run Cancel query Save query Export data as

```

1 SELECT * FROM Department;

```

Results Messages

Search to filter items...

DeptID	DeptName	Location
1	HR	New York
2	Finance	Chicago
3	IT	Boston
4	Marketing	Dallas

5.4 Blob Storage linked service and sink dataset

- Create a Blob Storage linked service.
- Select the appropriate authentication method.
- Create a DelimitedText dataset for the output.
- Choose your container, for example adfoutput.
- Enable First row as header if you want column headers in the CSV output.

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General
Connector upgrade advis...
Factory settings

Connections
Linked services
Integration runtimes
Microsoft Purview
ADF in Microsoft Fabric

Source control
Git configuration
ARM template

Author
Triggers
Global parameters
Data flow libraries

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name Annotations : Any

No linked service expected to see results, try changing

Create link

New linked service

Data store Compute

Search

All Azure Database File Generic protoc

Amazon RDS for Oracle Amazon RDS for SQL Server Amazon Redshift

Amazon S3 Amazon S3 Compatible Apache Impala

Microsoft Azure | Data Factory | datafactnan666

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Migrate to Fabric (Preview)

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name Annotations : Any

Showing 1 - 1 of 1 items

Name	Type	Related	Annotations
linked servicenans	Azure SQL Database	0	

Microsoft Azure | Upgrade

Search resources, services, and docs (G+/I)

Copilot

Home > Resource Manager | Resource groups > NANDINILDE > storeaccountnan666

storeaccountnan666 | Containers

Storage account

con

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix

Showing all 1 items

Name	Last modified	Anonymous access level
logs	8/25/2026, 2:18:21 PM	Private

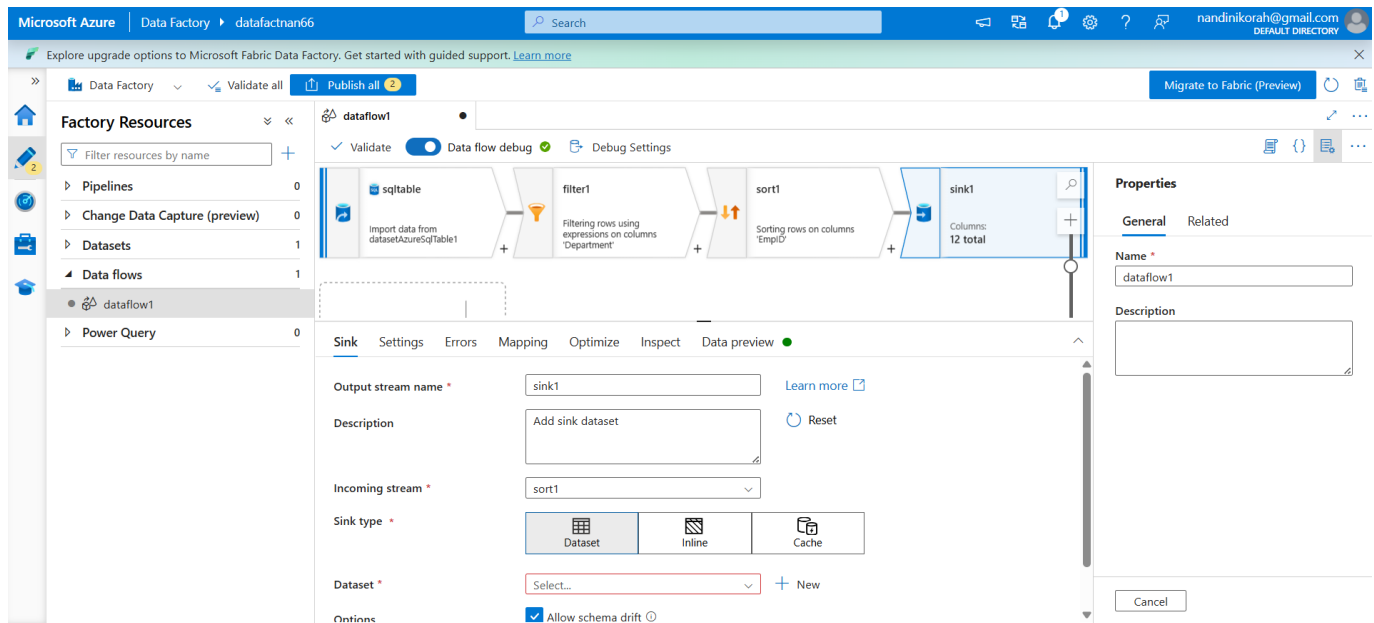
New container

Name *
output

Anonymous access level
Private (no anonymous access)

The access level is set to private because anonymous access is disabled on this storage account.

Advanced



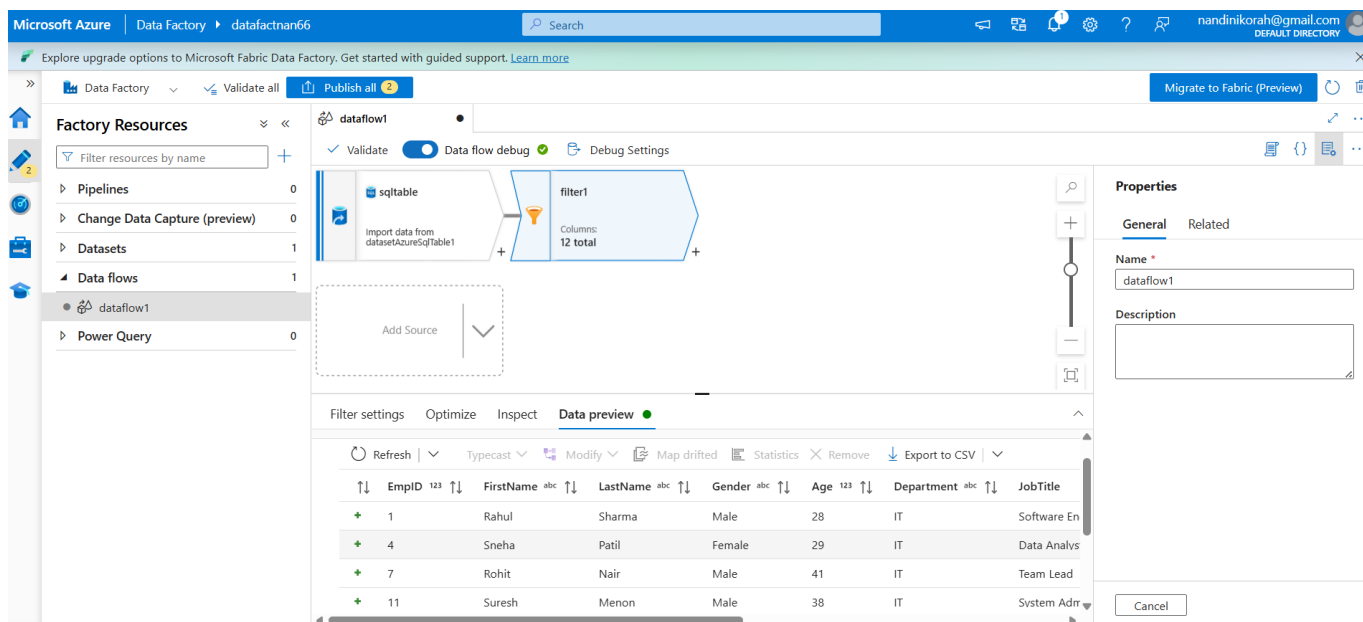
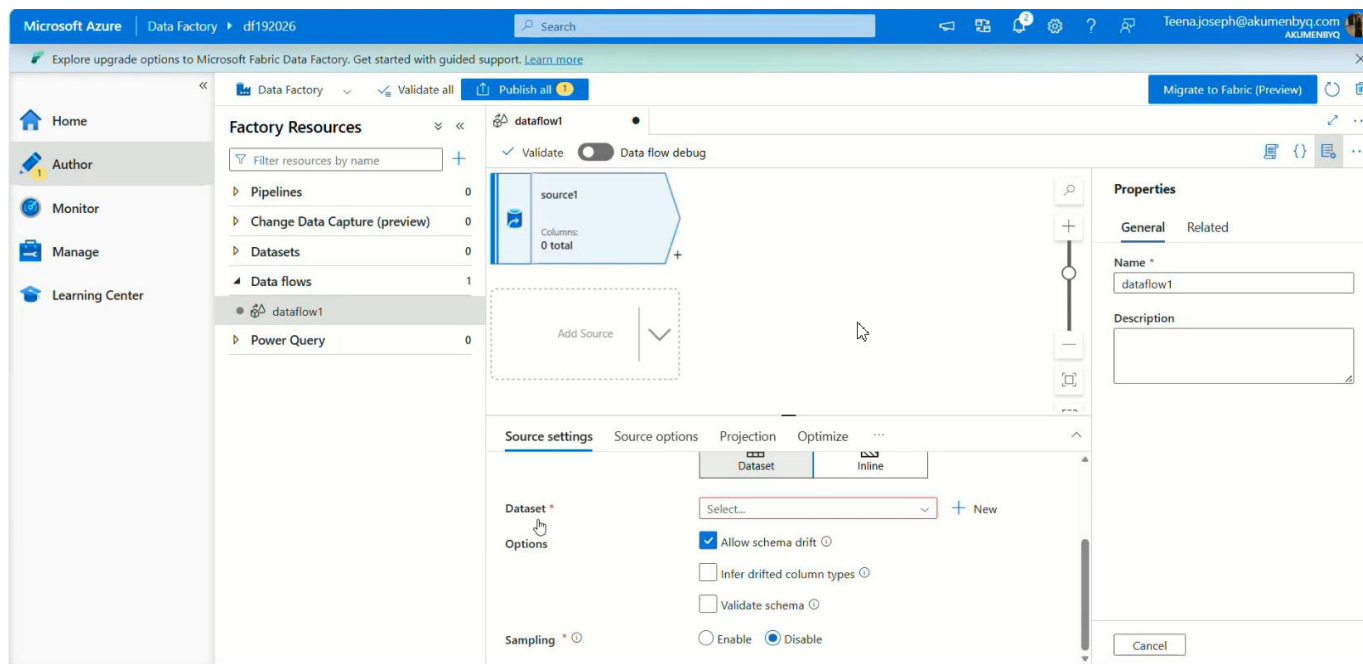
6. Data Flow 1 — EmployeeData → Filter → Sort → Blob Storage

This is the main transformation exercise. The source is the EmployeeData table in Azure SQL Database, and the final sink is a CSV file in Azure Blob Storage.

6.1 Create the Mapping Data Flow

- Go to Author → Data flows → + New data flow.
- Choose Mapping Data Flow.
- Name it DF_Employee_Filter_Sort.
- Turn Data Flow Debug on when you are ready to preview data. The debug cluster may take several minutes to start.

Microsoft's current documentation describes Mapping Data Flow as a visual transformation environment and recommends starting with a Source and ending with a Sink.



6.2 Add EmployeeData as Source

- On the canvas select Add Source.
- Name the stream EmployeeSource.
- Select DS_EmployeeData.
- Confirm that the projection contains the EmployeeData columns.
- Use Data Preview after Debug is active to confirm rows are being read.

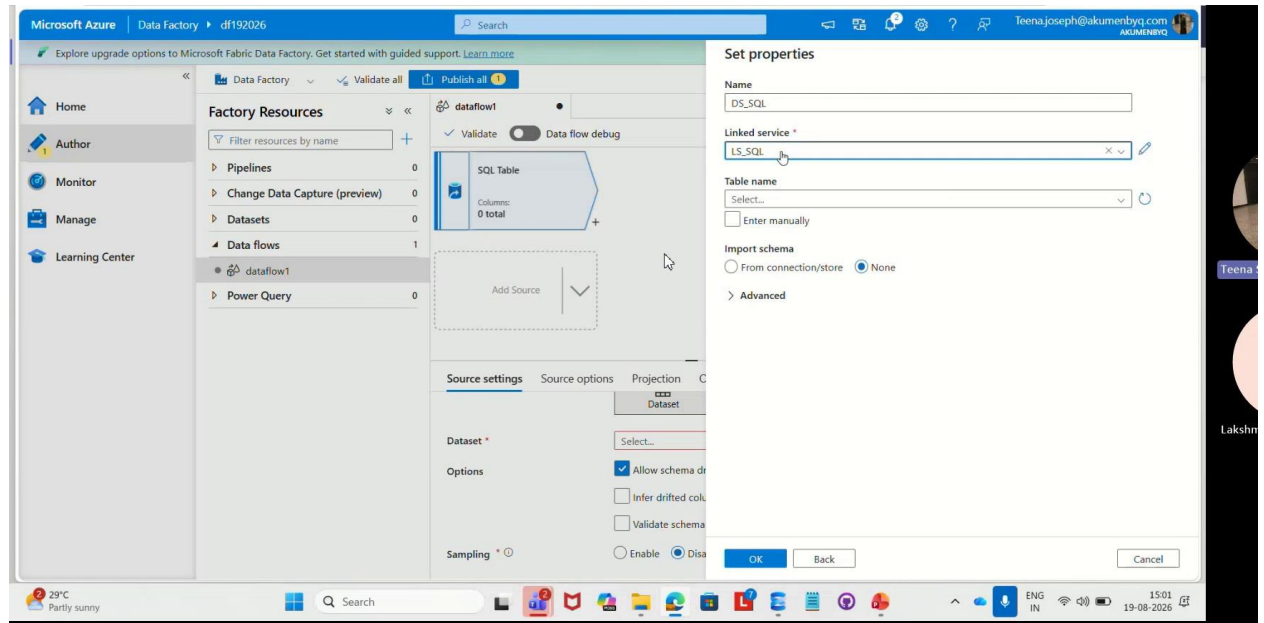


Figure 7 — Your recording: SQL source configuration used as the model for EmployeeData.

6.3 Add the Filter transformation

Filter keeps only rows that satisfy a Boolean expression. Conceptually, it performs the same job as a SQL WHERE condition.

- Select the + icon after EmployeeSource.
- Choose Filter.
- Name it FilterEmployees.
- In Filter on, open the expression builder.
- Enter a condition appropriate for your data.

Example: keep employees whose department is 'IT':

Department = 'IT'

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Factory Resources

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1
- Data flows 1
 - dataflow1
- Power Query 0

dataflow1

Validate Data flow debug Debug Settings

sqltable Import data from datasetAzureSqlTable1

filter1 Columns: 12 total

Add Source

Filter settings Optimize Inspect Data preview

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

EmpID	FirstName	LastName	Gender	Age	Department	JobTitle
1	Rahul	Sharma	Male	28	IT	Software En
4	Sneha	Patil	Female	29	IT	Data Analys
7	Rohit	Nair	Male	41	IT	Team Lead
11	Suresh	Menon	Male	38	IT	System Adm

Properties

General Related

Name * dataflow1

Description

Cancel

6.4 Add the Sort transformation

Sort orders the incoming rows by one or more columns. You can choose ascending or descending order.

- Select the + icon after FilterEmployees.
- Choose Sort.
- Name it SortBySalary.
- Select Salary as the sort column.
- Choose Descending if you want the highest-paid employees first.
- Add a second sort column such as EmployeeName if required.

Example:

Salary → Descending

EmployeeName → Ascending

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Factory Resources

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1
- Data flows 1
 - dataflow1
- Power Query 0

dataflow1

Validate Data flow debug Debug Settings

sqltable Import data from datasetAzureSqlTable1

filter1 Filtering rows using expressions on columns 'Department'

sort1 Columns: 12 total

Sort settings Optimize Inspect Data preview

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

Number of rows: INSERT 9 UPDATE 0 DELETE 0 UPSERT 0

EmpID	FirstName	LastName	Gender	Age	Department	JobTitle
1	Rahul	Sharma	Male	28	IT	Software En
4	Sneha	Patil	Female	29	IT	Data Analys
7	Rohit	Nair	Male	41	IT	Team Lead
11	Suresh	Menon	Male	38	IT	System Adm
14	Kavya	Shah	Female	28	IT	Backend De

Properties

General Related

Name * dataflow1

Description

Cancel

6.5 Add the Blob Storage Sink

- Select the + icon after SortBySalary.
- Choose Sink.
- Name it BlobEmployeeOutput.
- Select the Blob Storage dataset.
- Choose the output container/folder.
- Choose CSV/DelimitedText format.
- Enable First row as header.
- Review column mappings.
- Use Data Preview before running the complete pipeline.

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the Microsoft Azure logo, an 'Upgrade' button, a search bar, and a Copilot icon. Below the navigation bar, the user is logged in as 'nandinikoral'. The main content area shows the 'Overview' tab for a deployment named 'storeaccountnan666_1787647671494'. The deployment is in progress, with a start time of 8/25/2026, 2:17:56 PM and a correlation ID of f5b4368d-5d68-4edb-823c-2882c1dc2b15. A table shows the deployment details for the resource 'storeaccountnan666' of type 'Microsoft.Storage/storageAccounts', which is in an 'Accepted' status. On the right, there are links for 'Microsoft Defender for C', 'Free Microsoft tutorials', and 'Work with an expert'.

Home

storeaccountnan666_1787647671494 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name: storeaccountnan666_1787647671494
Subscription: Azure subscription 1
Resource group: NANDINI.DE

Start time: 8/25/2026, 2:17:56 PM
Correlation ID: f5b4368d-5d68-4edb-823c-2882c1dc2b15

Deployment details

Resource	Type	Status	Operation details
storeaccountnan666	Microsoft.Storage/storageAccounts	Accepted	Operation details

Give feedback

[Tell us about your experience with deployment](#)

Microsoft Defender for C
Secure your apps and infr
Go to Microsoft Defender

Free Microsoft tutorials
Start learning today >

Work with an expert
Azure experts are service
who can help manage you
and be your first line of su
Find an Azure expert >

Add or remove favorites by pressing Ctrl+Shift+F

The screenshot shows the Microsoft Azure Data Factory interface. The top navigation bar includes the Microsoft Azure logo, the Data Factory name 'datafactnan66', a search bar, and a user profile icon. Below the navigation bar, there's a 'Factory Resources' sidebar on the left with a search bar and a list of resources: Pipelines (1), Change Data Capture (preview) (0), Datasets (5), Data flows (2), and Power Query (0). The main canvas shows a pipeline named 'pipeline1' with a 'Dataflow1' tab selected. The pipeline consists of three dataflows: 'dataflow2' (Import data from dataset1AzureSqlTable1), 'join1' (Left outer join on 'dataflow2' and 'dataflow3'), and 'sink1' (Columns: 15 total). The 'sink1' dataflow is highlighted, and its properties are shown on the right. The properties include the name 'datafactsetdepartment', a description field, and a 'Sink type' dropdown set to 'Dataset'. The 'Dataset' dropdown is set to 'DelimitedText1'. The 'Options' section has a checkbox for 'Allow schema drift' which is checked. The bottom of the interface shows tabs for 'Sink', 'Settings', 'Errors', 'Mapping', 'Optimize', 'Inspect', and 'Data preview'.

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Data Factory | Validate all | Publish all

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines 1

pipeline1

Change Data Capture (preview) 0

Datasets 5

Data flows 2

dataflow1

datafactsetdepartment

Power Query 0

pipeline1

dataflow1

dataflow2

Import data from dataset1AzureSqlTable1

join1

Left outer join on 'dataflow2' and 'dataflow3'

sink1

Columns: 15 total

Properties

General Related

Name *

datafactsetdepartment

Description

Sink type *

Dataset

DelimitedText1

Test connection Open New

Skip line count

Options

Allow schema drift

Cancel

Microsoft Azure | Data Factory | datafactnan66

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Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 2
- Data flows: 1
 - dataflow1
- Power Query: 0

dataflow1

Validate Data flow debug Debug Settings

sqltable Import data from datasetAzureSqlTable1

filter1 Filtering rows using expressions on columns 'Department'

sort1 Sorting rows on columns 'EmpID'

sink1 Columns: 12 total

Properties

General Related

Name * dataflow1

Description

Number of rows INSERT N/A UPDATE N/A DELETE N/A UPSERT N/A

Refresh Statistics Export to CSV

EmpID	abc	FirstName	abc	LastName	abc	Gender	abc	Age	abc	Department	abc	JobTitle	abc
1		Rahul		Sharma		Male		28		IT		Software Engineer	
4		Sneha		Patil		Female		29		IT		Data Analyst	
7		Rohit		Nair		Male		41		IT		Team Lead	
11		Suresh		Menon		Male		38		IT		System Admin	
14		Kavya		Shah		Female		28		IT		Backend Developer	

Cancel

Microsoft Azure | Data Factory | datafactnan66

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Factory Resources

- Pipelines: 1
 - pipeline1
- Change Data Capture (preview): 0
- Datasets: 5
- Data flows: 2
 - dataflow1
 - dataflow2
 - dataflow3
- Power Query: 0

pipeline1

Validate Data flow debug Debug Settings

dataflow2 Import data from dataset1AzureSqlTable1

dataflow3 Import data from datasetAzureSqlTable2

join1 Left outer join on 'dataflow2' and 'dataflow3'

sink1 Columns: 15 total

Mapping

Options

- ☒ Skip duplicate input columns
- ☒ Skip duplicate output columns
- ☒ Auto mapping
- ☐ Reset
-
-
-

1 mappings: All inputs mapped

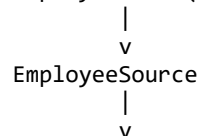
All inputs mapped by name including drifted columns

Microsoft Store

Figure 8 — Your recording: completed Azure Storage account that can be used for the Blob sink.

6.6 Final Data Flow 1

EmployeeData (Azure SQL)



```

FilterEmployees
  Department = 'IT'
  |
  v
SortByEmpID
  Empid asce
  |
  v
BlobEmployeeOutput
  Azure Blob Storage / CSV

```

7. Data Flow 2 — Join EmployeeData and Department

The Join transformation combines two input streams. For this exercise, EmployeeData supplies employee details and Department supplies the department name. Both tables are connected using DepartmentID.

7.1 Add the two sources

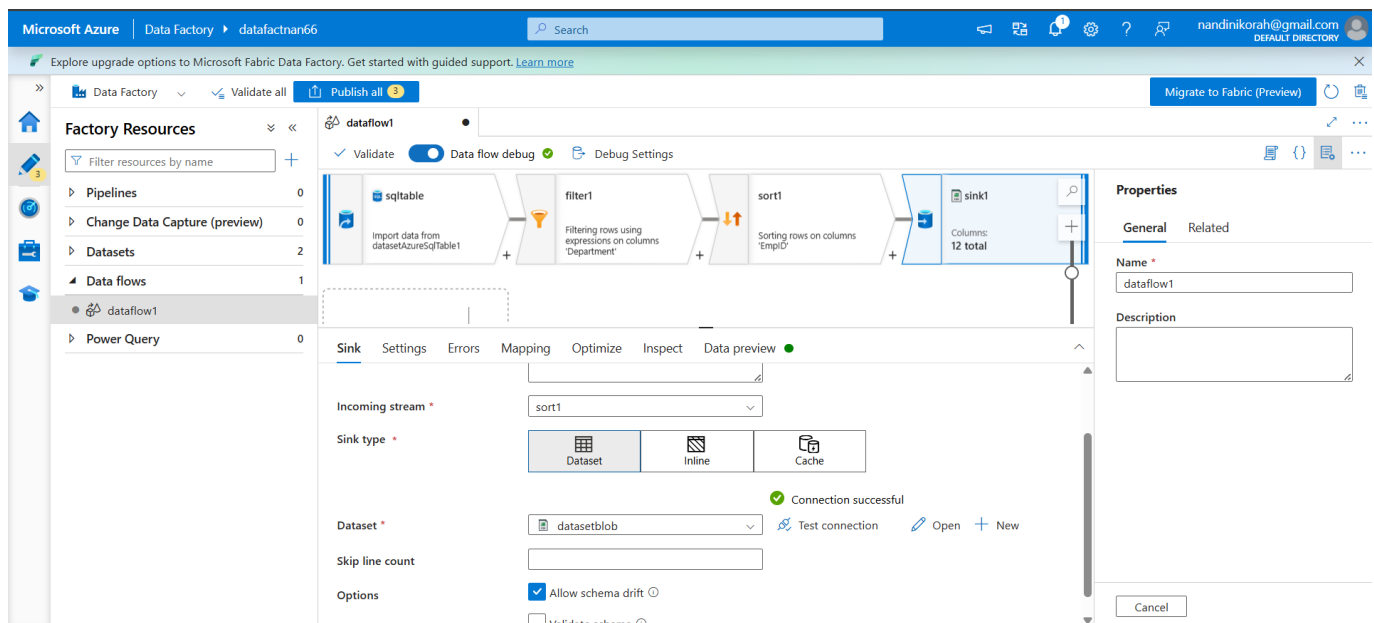
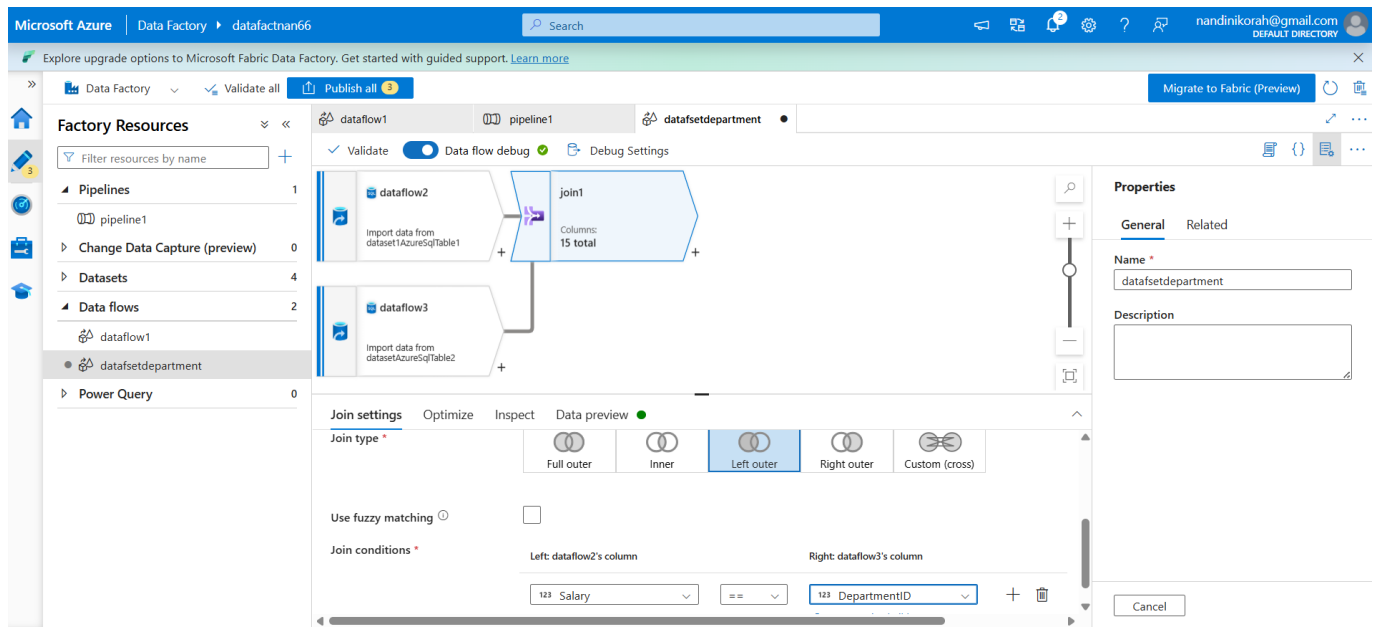
- Create a new Mapping Data Flow named DF_Employee_Department_Join.
- Add EmployeeData as the left source. Name it EmployeeSource.
- Add Department as the right source. Name it DepartmentSource.
- Make sure DepartmentID exists in both streams and has compatible data types.

A simplified source layout is:

EmployeeSource	DepartmentSource
EmployeeID	DepartmentID
EmployeeName	DepartmentName
DepartmentID	
Salary	

7.2 Add the Join transformation

- Select the + icon on EmployeeSource.
- Choose Join.
- Set EmployeeSource as the Left stream.
- Set DepartmentSource as the Right stream.
- Select left join for Join type if you only want employees with a matching department.
- Under Join conditions, select EmployeeSource.DepartmentID on the left and DepartmentSource.DepartmentID on the right.
- Keep the equality operator (==).
- Open Inspect/Data Preview to verify the combined columns.



7.3 Example join result

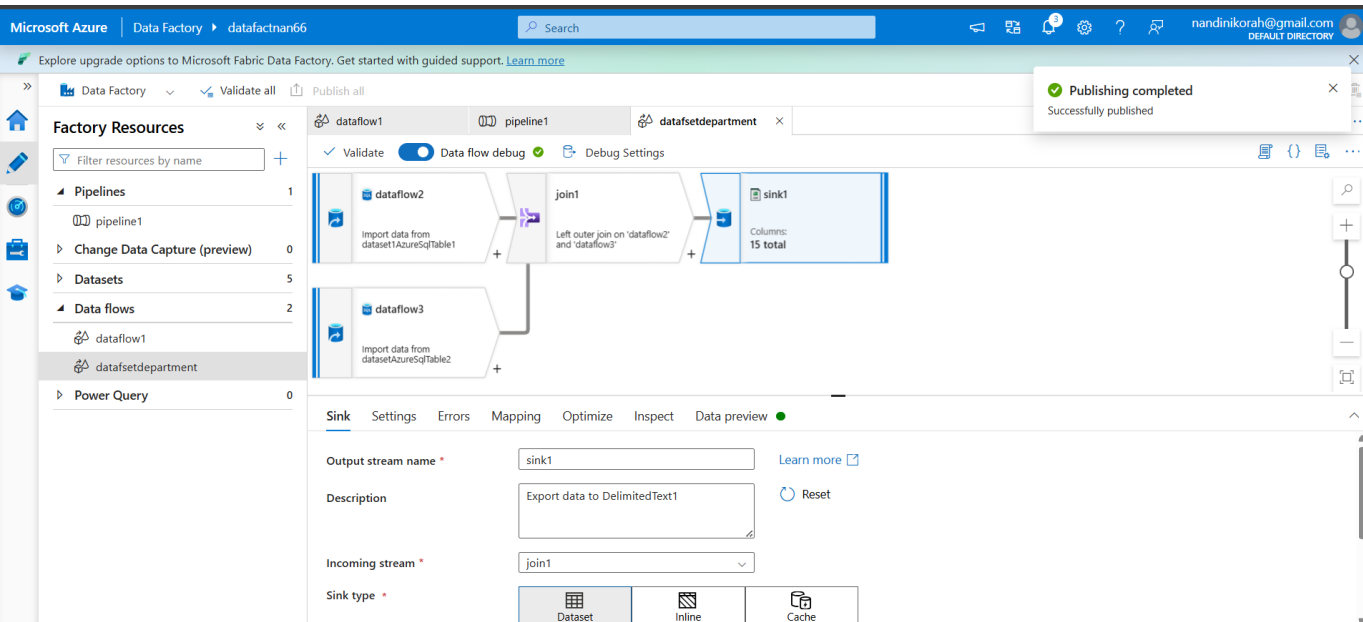
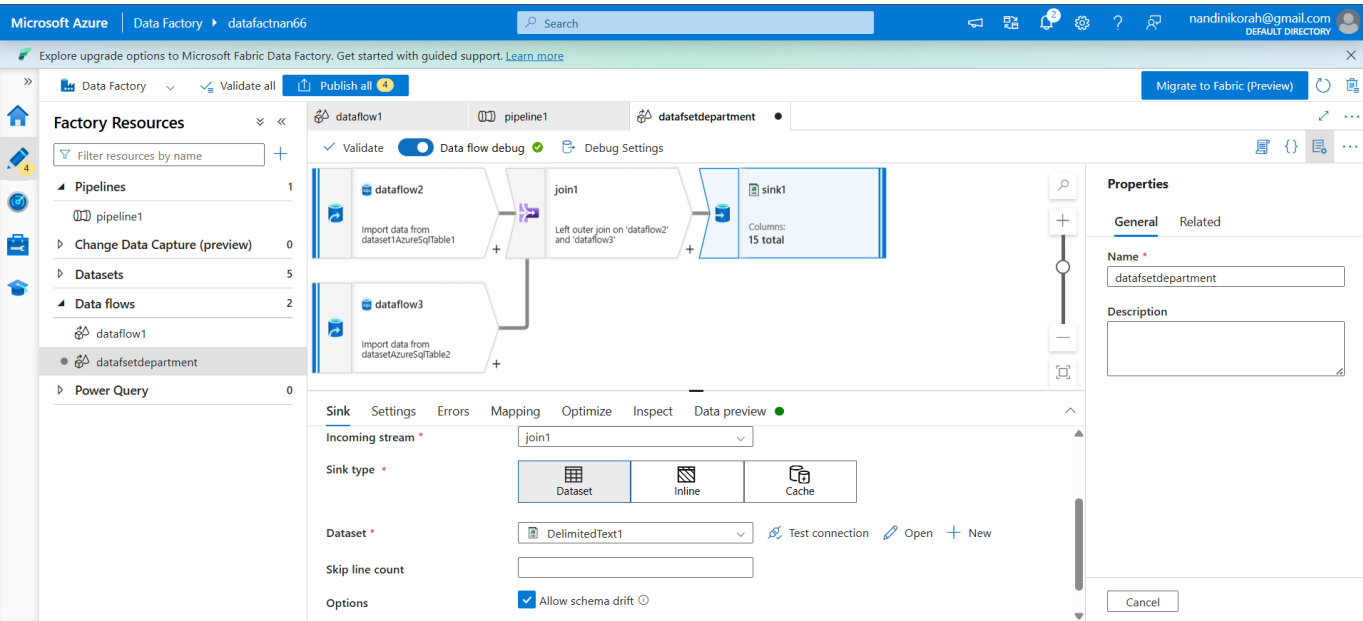
EmployeeID	EmployeeName	DepartmentID	Salary	DepartmentName
101	Anu	10	65000	IT
102	Rahul	20	58000	HR
103	Meera	10	72000	IT

7.4 Add a Select transformation after Join

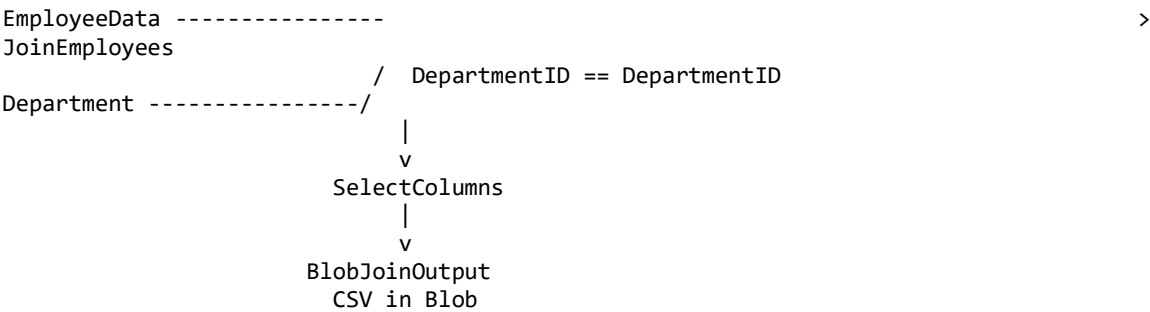
The Join output can contain duplicate key columns or columns that you do not need. Use Select to keep, rename or reorder columns.

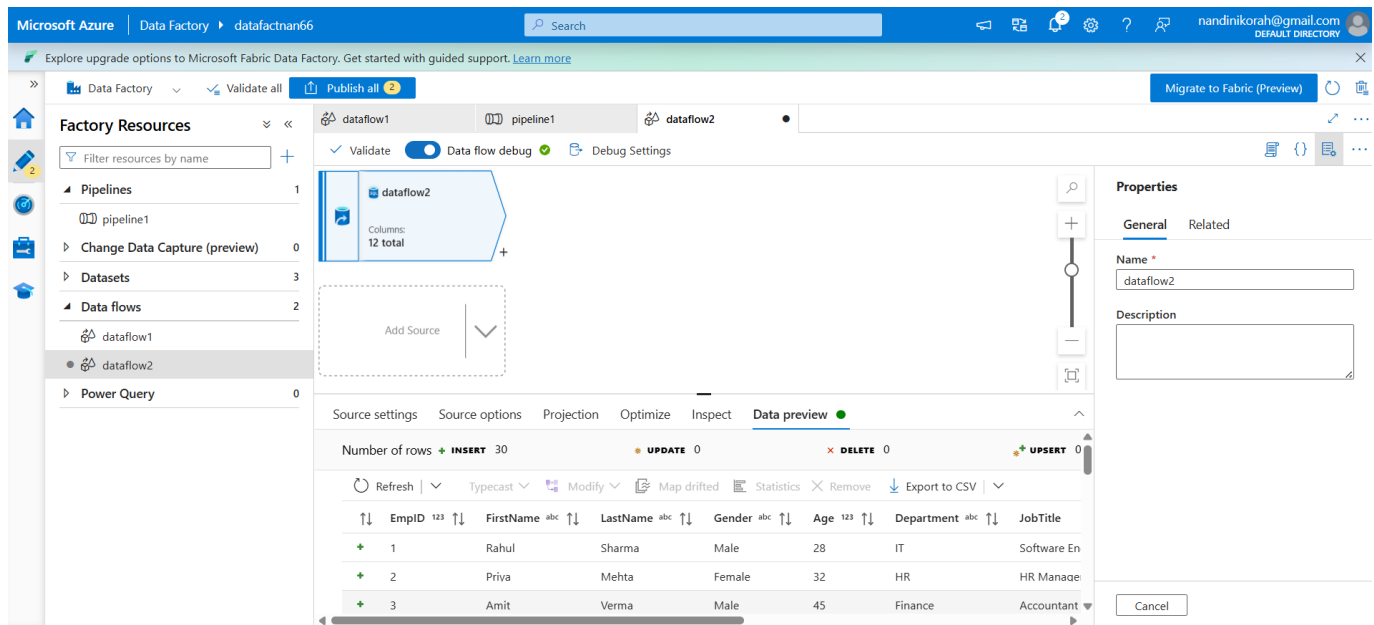
- Add Select after the Join.
- Keep EmployeeID, EmployeeName, Salary and DepartmentName.
- Optionally rename DepartmentName or reorder the columns.

- Then connect Select to the Blob Storage Sink.



7.5 Final Data Flow 2





8. Understanding ADF Join types

Join type	What it returns	Use in this project
Inner	Only matching rows from both EmployeeData and Department.	Recommended when every output employee must have a department.
Left Outer	All EmployeeData rows plus matching Department rows; unmatched department fields become NULL.	Useful when you want to keep every employee even if DepartmentID is missing.
Right Outer	All Department rows plus matching employee rows.	Useful for auditing departments and identifying departments with no employees.
Full Outer	All rows from both streams; unmatched columns become NULL.	Useful for reconciliation and finding unmatched records on both sides.
Custom Cross	Rows based on a custom/non-equality condition.	Use only when a normal equality join is not sufficient.

Microsoft Learn currently documents five mapping data flow join types: Inner, Left Outer, Right Outer, Full Outer and Custom Cross. The Join transformation matches rows according to the configured join condition.

9. Inner Join vs Left Outer Join — simple example

EmployeeData

```
101 Anu 10
102 Rahul 20
103 Meera 30
```

Department

```
10 IT
20 HR
```

INNER JOIN result

```
101 Anu 10 IT
102 Rahul 20 HR
```

```
LEFT OUTER JOIN result
101 Anu      10 IT
102 Rahul    20 HR
103 Meera    30 NULL
```

10. Running the pipeline

- Create a Pipeline under Author.
- Add a Data Flow activity under Move & Transform.
- Select the required Mapping Data Flow.
- Choose the Integration Runtime if prompted.
- Click Debug to test the pipeline.
- Check Output/Monitor for row counts and errors.
- After successful testing, Validate the pipeline.
- Publish all changes.
- Use a manual trigger or schedule trigger when ready.

Official reference: [Create a Mapping Data Flow — Microsoft Learn](#)

11. Data Preview and debugging

- Turn on Data Flow Debug before using Data Preview.
- Open Data Preview on the source to confirm EmployeeData is being read.
- Open Data Preview after Filter to confirm only matching rows remain.
- Open Data Preview after Sort to check the order.
- Open Data Preview after Join to confirm DepartmentName is populated.
- If the Join returns zero rows, check DepartmentID values and data types on both sides.
- If the Blob file is empty, check the sink dataset, folder path, permissions and column mappings.

12. Common problems and fixes

EmployeeData does not appear in the dataset table list: Refresh the dataset/table metadata, verify the SQL linked service connection, confirm the table exists in the selected database/schema, and verify that the login has permission to read it.

Join returns no rows: Check that DepartmentID exists in both streams, has compatible data types, and contains matching values.

Duplicate DepartmentID columns after Join: Use Select after Join to keep only the required key column and business columns.

Blob output is not created: Check the Blob linked service, container/folder path, sink dataset format and permissions. Then run Debug and inspect the activity output.

Sort appears not to be preserved: Mapping Data Flow runs on distributed Spark partitions. If global ordering is important, keep Sort close to the Sink and review the Optimize settings.

13. Recommended project flow

PROJECT 1 – FILTER + SORT
Azure SQL EmployeeData
↓
ADF Mapping Data Flow
↓
Filter: Department = 'IT'
↓
Sort: Salary DESC
↓
Blob Storage: EmployeeData_Filtered_Sorted.csv

PROJECT 2 – JOIN
Azure SQL EmployeeData
↓
Join on DepartmentID
↓
Select columns
↓
Blob Storage: Employee_Department.csv

14. Key points to remember

- Copy Activity is primarily for moving data; Mapping Data Flow is the visual option to perform transformations such as Filter, Sort and Join.
- Filter is similar to a SQL WHERE condition.
- Sort orders rows by selected columns.
- Join combines two streams using matching or custom conditions.
- EmployeeData.DepartmentID = Department.DepartmentID is the natural join condition for this example.
- Use Select after Join to control the final output schema.
- Use Data Preview and Debug to test each transformation before publishing.
- For large joins, review broadcast/partitioning options and avoid unnecessary cross joins.